Glossary

absolute space: Newton’s view of space; envisions space as unchanging and independent of its contents.

absolute spacetime: View of space emerging from special relativity; envisions space through the entirety of time, from any perspective, as unchanging and independent of its contents.

absolutist: Perspective holding that space is absolute.

acceleration: Motion that involves a change in speed and/or direction.

accelerator, atom smasher: Research tool of particle physics that collides particles together at high speed.

aether, luminiferous aether: Hypothetical substance filling space that provides the medium for light to propogate; discredited.

arrow of time: Direction in which time seems to point—from past to future.

background independence: Property of a physical theory in which space and time emerge from a more fundamental concept, rather than being inserted axiomatically.

big bang theory/standard big bang theory: Theory describing a hot, expanding universe from a moment after its birth.

big crunch: One possible end to the universe, analogous to a reverse of the big bang in which space collapses in on itself.

black hole: An object whose immense gravitational field traps anything, even light, that gets too close (closer than the black hole’s event horizon).

braneworld scenario: Possibility within string/M-theory that our familiar three-spatial dimensions are a three-brane.

Casimir force: Quantum mechanical force exerted by an imbalance of vacuum field fluctuations.

classical physics: As used in this book, the physical laws of Newton and Maxwell. More generally, often used to refer to all nonquantum laws of physics, including special and general relativity.

closed strings: Filaments of energy in string theory, in the shape of loops.

collapse of probability wave, collapse of wavefunction: Hypothetical development in which a probability wave (a wavefunction) goes from a spread-out to a spiked shape.

Copenhagen interpretation: Interpretation of quantum mechanics that envisions large objects as being subject to classical laws and small objects as being subject to quantum laws.